# UNIT 2: SHAPES AND REGIONS

## SESSION 4: BOUNDARY TRACKING PROCEDURES

## SLO 1: UNDERSTAND AND IMPLEMENT BOUNDARY TRACKING TECHNIQUES TO EXTRACT OBJECT CONTOURS IN BINARY IMAGES

**Think and Apply**

**Consider two objects in a binary image.**

| Object | Shape |
|---|---|
| Object A | Square |
| Object B | Circle |

**Questions**

1. **Which object will have a longer boundary?**

   **Ans- Object A (Square) will have a longer boundary, assuming both objects are of comparable size.**

2. **How can boundary tracking help in shape recognition?**

   **Ans- Boundary tracking helps in shape recognition by extracting the contour or outline of an object. The tracked boundary provides information about the object's shape, size, and structure, making it easier to distinguish between different shapes such as squares, circles, and other objects.**

**Mini Challenge**

**Why is boundary tracking useful in applications like object detection, medical image analysis, and character recognition?**

**Ans- Boundary tracking is useful in object detection, medical image analysis, and character recognition because it extracts the outer contour of objects and makes their shape easier to analyze. In object detection, it helps identify and separate objects from the background. In medical image analysis, it can be used to identify the boundaries of organs, cells, or abnormal regions. In character recognition, the contours of letters and numbers help the system recognize their shapes. Thus, boundary tracking simplifies image analysis by focusing on the important shape information.**

**Reflection**

**Write one advantage of extracting object contours instead of analyzing all pixels in the object.**

**Ans- One advantage of extracting object contours is that it reduces the amount of data that needs to be analyzed. Instead of processing every pixel inside an object, the system mainly analyzes the boundary, which contains important information about the object's shape.**